

B.Tech. Degree II Semester Regular Examination July 2024

CE/ME/SE 23-200-0204A BASIC ELECTRICAL ENGINEERING

(2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Explain the concepts of various energy sources and electric circuits.
 CO2: Apply the basic electrical laws to solve circuits.
 CO3: Discuss the construction and operation of various electrical machines.
 CO4: Identify suitable electrical machine for practical implementation.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

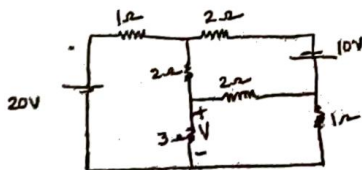
(Answer **ALL** questions)

	(5 × 2 = 10)	Marks	BL	CO	PO
I. (a) Define		2	L1	1	1
(i) Self induced emf.					
(ii) Mutual induced emf.					
(iii) Coefficient of coupling.					
(b) Prove that the power drawn by a pure inductor over a complete cycle is zero.		2	L2	1	1
(c) Derive the expressions for impedance, power factor and current of a RLC series circuit supplied by a sinusoidal voltage source. Draw the vector diagram.		2	L2	2	1
(d) Develop from first principles, an expression for the emf generated in DC generator.		2	L2	3	1
(e) Write short notes on		2	L1	4	2
(i) Fuse.					
(ii) MCB.					

PART B

(4 × 10 = 40)

II. (a)	A circuit consisting of three resistors 12 ohm, 18 ohm and 36 ohm in parallel and the combination connected in series with a fourth resistor R is supplied with a potential difference of 60 V. It is found that the power dissipated in the 12 ohm resistor is 36 W. Determine the value of R and the power dissipated in it.	5	L2	1	1
(b)	Explain hydroelectric power generation with the help of neat sketch	5	L1	1	1
OR					
III. (a)	Determine the voltage V across the resistor 3 ohm.	6	L3	1	2



(b)	State Faraday's law and Lenz's law of electromagnetic induction.	4	L1	1	2
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(P.T.O.)

		Marks	BL	CO	PO
IV. (a)	Determine the resistance and reactance of the element if the current through the element is $10 \sin(\omega t - 30^\circ)$ and the voltage across the element is $100 \sin(\omega t + 30^\circ)$. Also find the real power, reactive power and apparent power drawn by the element.	5	L2	2	2
(b)	A coil of resistance 10 ohm, inductance 1H and capacitance $15.83 \mu\text{F}$ are connected in series across a 100 V supply. The current drawn is 10 A. Determine the frequency of the supply.	5	L3	2	2
OR					
V. (a)	A 3 phase load has a resistance of 10 ohm in each phase against a 400 V, 3 phase supply and is connected in (i) star (ii) delta Compare the power consumed in both cases.	6	L1	1	2
(b)	Define rms value and derive the expression for rms value of a sine wave.	4	L1	1	2
VI. (a)	A 110 V, DC shunt generator delivers a load current of 50 A. The armature resistance is 0.2 ohm and the field resistance is 55 ohm. The generator rotating at a speed of 1800 rpm has 6 poles, lap wound and a total of 360 conductors. Calculate the flux per pole.	6	L1	3	1
(b)	Compare the electrical and mechanical characteristics of dc series and shunt motors.	4	L1	3	1
OR					
VII. (a)	A 4 pole, 500 V, 74.6 KW wave connected motor has 492 conductors. The flux per pole is 50 mWb and full load efficiency is 92%. Its shunt field resistance is 250 ohm and armature resistance is 0.1 ohm. Calculate the motor speed for full load.	6	L2	3	1
(b)	Explain the classification of dc generators based on its field excitation.	4	L3	3	2
VIII. (a)	A 200 kVA, 3300/24V, 50 Hz single phase transformer has 80 turns on the secondary winding. Assuming as an ideal transformer, calculate (i) full load secondary current. (ii) maximum value of flux. (iii) number of primary turns.	6	L2	4	2
(b)	Distinguish the types of transformer and explain the elementary theory of an ideal transformer.	4	L1	4	1
OR					
IX. (a)	A 50 kVA single phase transformer draws a primary current of 250 A on full load. The total resistance referred to primary side is 0.006 ohm. If the iron loss of the transformer is 200 W, calculate the efficiency on full load and on half load at 0.8 pf lagging.	6	L3	4	2
(b)	Write the classification of three phase induction motor based on its construction. Which induction motor is most commonly used for industrial applications? Why?	4	L2	4	2

Blooms's Taxonomy Levels

L1 - 30%, L2 - 45.8%, L3 - 24.2%.
